Nitish Agarwal

agarwalnitish121@gmail.com
+91-7011690146
https://www.linkedin.com/in/agar-nit/
https://github.com/agarNit/
https://leetcode.com/u/agar_Nit
https://agarnit.github.io

Backend Software Engineer with 2+ years of experience building production-scale distributed systems and event-driven architectures on AWS. Cut infrastructure costs by 50% while improving system reliability to 99.9% uptime. Skilled in microservices, Kafka streaming, and real-time data pipelines supporting 5K+ concurrent users.

WORK EXPERIENCE

Korefi.ai Aug 2026 – Present
Software Development Engineer II Bengaluru, Karnataka

- Launched a **financial-operations MVP** for **US restaurants**, now serving 19 brands across 37 outlets. It tracks $1.66M in delivery sales (57K orders) across DoorDash, Uber Eats and Grubhub, unifies **POS and P&L** data, and surfaces chargebacks, recovering about 80% in platform errors.
- Engineered a **multi-stage AI pricing analyst** that benchmarks restaurant menus against nearby competitors. It uses LLM cuisine detection and competitor ranking, plus a **two-model dish-matching layer** with swapped-pair verification and cached verdicts. A browser-agent fallback reads hard-to-scrape menus, and it has priced 174 competitors and 157K menu items till now.
- Architected "**Benji**", a tool-calling accounting agent for **QuickBooks** on the **Vercel AI SDK**, with 48 tools and human-approved writes only. It has idempotent, undoable posting, validated journals, a step-capped loop, and evaluation with tracing. It also has persistent conversation memory and chart-of-accounts context, and has been piloted on 3 QuickBooks companies.
- Automated sales-report retrieval across 3 delivery portals for 19 restaurant brands through a **Microsoft Teams bot**. It uses per-brand browser sessions with **encrypted logins** and human-in-the-loop verification. It prevents duplicate uploads and feeds the ingestion pipeline automatically.
- Working across a Python stack - FastAPI, Django, PostgreSQL/Supabase, AWS and Docker, with Langfuse for LLM tracing, prompt evaluation, and observability.

Spectatr.ai Jul 2024 – Jul 2026
Software Development Engineer I Gurugram, India

- Architected an **automated video streaming system** handling 50+ live streams simultaneously, cutting delay from upload to playback to under 8 seconds and eliminating all duplicate processing errors (previously 15–20 daily) using **smart caching**.
- Achieved 99.9% **reliable message delivery** (zero data loss across 20K+ daily events) by building scalable streaming data pipeline that handles real-time analytics for 10K+ simultaneous viewers.
- **Reduced inappropriate chatbot requests** by 17% (blocking 8,500+ monthly queries) through smart filtering system, saving ~₹45K monthly in processing costs and preventing inaccurate responses from reaching users.
- Improved chatbot accuracy from 68% to 94% (81% reduction in incorrect responses) by implementing **quality monitoring framework** with automated testing and continuous improvement, serving 15K+ daily user queries.
- Fixed **critical file deletion bug** causing 40+ daily data errors by implementing **queued processing with smart locking**, improving storage tracking accuracy from 26% to 99% and eliminating manual cleanup work.

PROJECTS

Cloudy
AI Coding Assistant CLI (*Claude Code Clone*)

- Built Cloudy, a **Claude Code-style AI coding assistant CLI** in Python using LangChain, LangGraph, and the Anthropic Claude API, enabling autonomous, multi-step, tool-calling agents to reason over any codebase.
- Engineered a **RAG pipeline** with AST-based code parsing (tree-sitter, 15+ languages) for structure-aware chunking, plus hybrid dense and sparse **vector retrieval** (Qdrant, sentence-transformers, BM25), with automatic fallback to embedded ChromaDB for zero-config use.
- Designed **agentic memory** and **safety systems**, including short-term and long-term memory with auto-summarization, session management, a three-tier progressive-disclosure skills framework, and **human-in-the-loop approval** gates for destructive actions.
- Implemented **production guardrails** including PII and secret redaction middleware, OS-level shell sandboxing (Docker and macOS Seatbelt), and per-tool output caps to prevent LLM context-window overflow, with automatic session recovery on failure.
- Built an **evaluation and observability stack**, including a LangSmith-sourced **golden dataset**, deterministic and LLM-judge scoring, a correction loop, and dual tracing (LangSmith and **Langfuse**) for cost and latency monitoring.
- Packaged and shipped the CLI for public **pip/pipx installation** with prompt and semantic caching and a self-updating release-notification system, then root-caused and fixed a live production incident reported by an external user across three layers of defense.

TECHNICAL COMPETENCIES

- **Languages & Frameworks:** TypeScript, Python (FastAPI), Java, Express.js, LangChain, LangGraph, Node.js, Golang
- **Backend & Systems:** MongoDB, Redis, DynamoDB, Kafka, AWS SQS, SNS, ECS, Lambda, MediaLive, S3, CloudFront, Docker, Kubernetes, Distributed Systems, Microservices, Event-Driven Architecture, REST APIs, System Design, CI/CD, Fault Tolerance
- **AI/ML and LLMs:** Multi-agent systems (Langgraph), Production LLM deployment, Prompt engineering, Hallucination detection, RAG pipelines, AWS Bedrock, Langsmith evaluation frameworks, Agentic AI

EDUCATION

Chandigarh University, Punjab Oct 2020 – Jun 2024
Bachelor of Engineering, Computer Science CGPA: 8.25

IIT Madras Nov 2020 – Dec 2023
Bachelor of Science, Data Science and Programming (*Online*) CGPA: 7.73